Silent-Failure Study

Status as of 25 August 2026 · **Week 1 complete**

Where you are: all three engines are built, validated, and compared.

No engine ever returned a wrong answer. The study's original claim — that engines fail silently — is so far unsupported. What was found instead is a clean capability difference and a well-evidenced null result.

Done

- ✓ **Engine frozen** at `a8dc2b2` / `v2.1.3`; baseline re-measured and **bit-identical** to the 10 August run (44 pass / 0 silent / 1 error / 10 timeout).
- ✓ **Independent referee built** — `reference.py`, closed form, `numpy` only, sharing no library with any engine. Eleven self-tests. Needed because a SymPy oracle cannot judge a SymPy engine.
- ✓ **Three engine adapters**: MechanicsDSL (native), `sympy.physics.mechanics` (idiomatic), Drake 1.56 (WSL).
- ✓ **Compared on derivation** — all three agree with the referee to $\leq 7 \times 10^{-14}$.
- ✓ **Compared on integration**, regular *and* chaotic regimes, one pinned integrator. Energy drift identical across engines.
- ✓ **Amplitude axis added** after measurement showed the suite had only tested the chain where it is well behaved (perturbations grew 2–22×; at 3 rad they grow $10^{11}\times$). 24 engine–amplitude pairs, zero failures.
- ✓ **Scaling walls measured** to 30 links — the study's first genuine disagreement.

The headline result

Largest chain each engine will answer:

	Lagr.	Hamil.
SymPy (idiomatic)	5	—
MechanicsDSL	12	3
Drake	**30**	—
referee	≥ 30	≥ 30

Drake's cost is *flat* at 0.23 s from 5 links to 30. The two symbolic engines slow down and then stop.

But every engine that answered, answered correctly, and every engine that could not simply never returned. Timing out is the *loud* failure mode — the acceptable one. The engines differ in reach, not in honesty.

Also found

- **!** At the inverted equilibrium ($\theta = \pi$) all three engines *and* the referee report success while producing ~20 rad of motion where the exact answer is none. Cause is $\sin \pi = 1.22 \times 10^{-16}$ in float64, amplified by an unstable equilibrium. Not an engine defect — but nobody warns you. A shared blind spot.
- **!** Two apparent silent failures were *my* coordinate-convention errors, not engine defects: the canonical (q, p) state, and Drake's relative joint angles. Both vanished at N=1 and grew with N — the signature of a convention mismatch. Check that pattern before reporting any disagreement.
- **!** Three study documents described the study inaccurately and were corrected: `SCOPE.md` claims one test family (there are three); the portable-axes list named a non-existent axis; the instrument-freeze rule forbade Week 1.

Rules still in force

- **!** **Engine, scoring path, and the 55-case matrix are frozen.** The harness may grow; the gate is that the frozen matrix still returns 44/0/1/10 bit-identically. Extending coverage is allowed, revising judgement is not.
- **!** **Quote numbers only from `RESULTS.md`**, and state them *by pathway* — the strong oracle covered 22 of 45 adjudicated cases. The 28 July report describes a pre-fix engine.
- **!** **Do not claim MechanicsDSL outperforms Drake.** Identical energy drift under a *pinned* integrator cannot rank engines, and a 0.2 % difference is noise. Reaching 12 links against idiomatic SymPy's 5 is defensible; Drake reaches 30.

The decision waiting for you

- ☐ **Report the null result.** Three independent engines, adjudicated by a fourth derivation sharing no library, agreeing to machine precision across an adversarial set spanning both dynamical regimes. Real, well supported, and a different paper from the one planned.
- ☐ **Or keep hunting** in the untested ground: the constrained pathway and the `loops` axis, neither yet compared across engines.

Still open either way: whether `loops` enters the cross-engine claim, and whether to re-express the two spring systems as pendulum chains so they become portable to a rigid-body engine.

Reports: TR-2026-01 referee · 02 SymPy · 03 Drake · 04 Week 1 findings. **Disclose:** you maintain one of the engines measured, and AI assistance was used. Both permitted; concealing either is not.